

# AI and Blockchain-Integrated Mock Interview and Collaboration Platform

**Abstract:** This paper proposes a secure, AI-driven web platform that allows users to register, build a professional profile, take skill-based mock interviews, connect with peers through online communities, and store their achievements securely on a blockchain ledger. The system integrates three core components — (1) an intelligent mock interview module that evaluates user performance based on their skills, (2) a collaborative community interface for networking and project-based learning, and (3) a blockchain layer for decentralized and tamper-proof achievement verification.

**Keywords:** Blockchain, Skill Assessment, Mock Interview, AI Evaluation, Decentralized Certification, Flask, React.js, Web3, Community Collaboration.

**1. Introduction** The growing importance of skill-based recruitment has highlighted the limitations of traditional interview systems. Many existing solutions fail to ensure transparency, data security, and real-time skill verification. This research introduces a unified platform that combines AI-based skill evaluation with blockchain-secured achievements to create a trustworthy and collaborative learning ecosystem.

**2. Literature Survey** Several AI-based learning and recruitment platforms exist, but most lack integration with blockchain for achievement verification. Previous works focus either on mock interviews or certificate issuance, but none provide a complete system combining assessment, collaboration, and secure credential storage.

**3. Problem Definition** Traditional interview systems lack automation and credibility. Certificates can be tampered with and are not easily verifiable. There is also a gap between learning platforms and collaboration environments. This project aims to build a system where users can take skill-based interviews, receive scores, collaborate with peers, and have their achievements securely recorded on blockchain.

**4. System Architecture** The proposed system includes modules for User Management, Mock Interview Engine, Community Collaboration, and Blockchain Integration. The frontend (React.js) interacts with the Flask backend through REST APIs. PostgreSQL stores user and interview data, while blockchain ensures secure and immutable achievement storage.

**5. Methodology** - User Module: Handles registration, login, and profile management. - Interview Module: Selects questions based on user skills, computes scores, and updates dashboards. - Community Module: Allows users to connect, share posts, and collaborate. - Blockchain Module: Stores hashes of user achievements using smart contracts deployed on Ethereum/Polygon. - Dashboard: Displays interview history, community updates, and verified achievements.

**6. Implementation and Results** The platform is implemented using Flask (backend) and React.js (frontend). PostgreSQL is used for data storage. Blockchain integration uses Solidity smart contracts and Web3.py for interaction. Initial testing showed secure storage of certificates and efficient retrieval of mock interview results.

**7. Security and Blockchain Integration** Blockchain ensures immutability and transparency. Achievements are stored as hashed data on-chain to protect user privacy while maintaining public verifiability. Smart contracts manage issuance and retrieval, ensuring decentralized verification of user credentials.

**8. Future Enhancements** - Add AI-driven voice interviews using NLP and speech recognition. - Enable multi-chain compatibility for global achievement verification. - Introduce recruiter login to shortlist candidates based on blockchain records. - Add mobile application support.

9. Conclusion This research presents a comprehensive system combining AI-based skill assessment, peer collaboration, and blockchain-secured achievements. The platform enhances employability by providing transparent, verifiable, and intelligent evaluation mechanisms.

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